

Software Testing (CS 4036)

Sessional-I Exam

Date: 24 September, 2025

Course Instructor(s):

Mr. M. Waqas

Total Time (Hrs): 1

Total Marks: 25

Total Questions: 2

Roll No

Section

Student Signature

Answer all questions. Answer questions in the same order as they appear on question paper.
Answer all questions on answer sheets.

Question1: Consider the following program and answer the questions.

```
void calculate_factorial(int n)
{
    int result = 1;

    if (n < 0)
    {
        std::cout << "Error: Factorial is not defined for negative numbers." << std::endl;
        return;
    }

    // C. Loop for calculation
    for (int i = 1; i <= n; i++)
    {
        result = result * i;
    }

    std::cout << "The factorial of " << n << " is: " << result << std::endl;
}
```

National University of Computer and Emerging Sciences

Lahore Campus

Exam questions.

1. Draw dataflow graph (definition and use nodes for variables clearly annotated). [10]
2. Identify paths to satisfy all DU path criteria w.r.t. to all variables. [3]
3. Find out DCF(definition clear path in each DU path.) [2]

Question2: Consider the following function code and control flow graph of check_leap_year. Answer the following questions accordingly.

```
void check_leap_year(int year) {  
    if (year % 4 == 0) {  
        if (year % 100 == 0) {  
            if (year % 400 == 0) {  
                std::cout << year << " is a leap year." << std::endl;  
            } else {  
                std::cout << year << " is not a leap year." << std::endl;  
            }  
        } else {  
            std::cout << year << " is a leap year." << std::endl;  
        }  
    } else {  
        std::cout << year << " is not a leap year." << std::endl;  
    }  
}
```

1. What is the minimum number of test cases required to perform basis path testing of the program? Show the complete process of finding the number of required test cases. [3]
2. Identify and list the basis paths. [5]
3. Provide at least 2 test cases. [2]

Note: CFG is provided on the next page.

